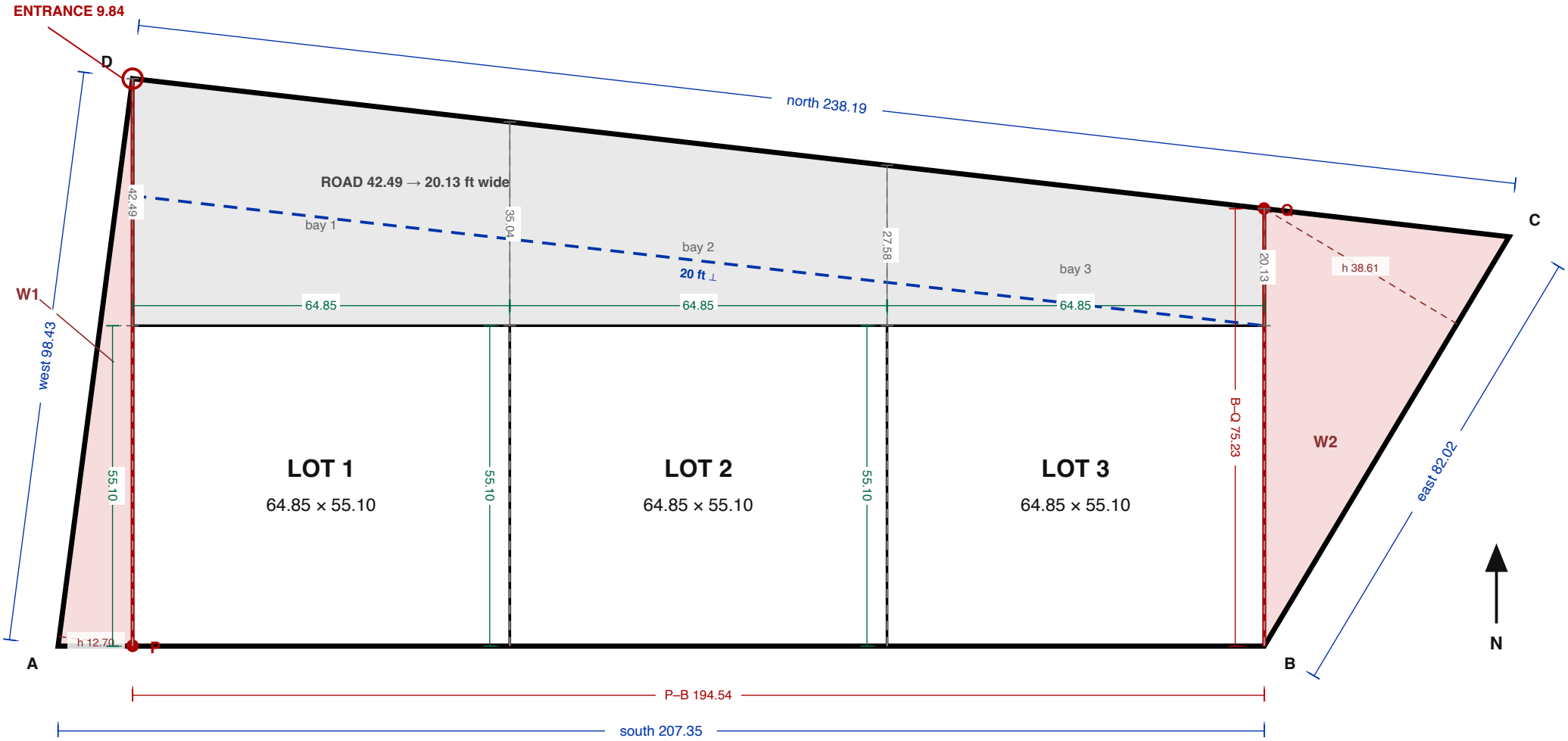
TS 11/14/18 · Ward-16 Thallakulam · Madurai North

OPTION A — LEVEL

scale 1:300 — print at 100% on A4

north up · all figures in feet



SCALE 1:300 — 1 mm on paper = 0.30 m = 0.98 ft on the ground

0 5 10 15 20 25 m

this bar is exactly 100 mm — check it with a ruler; if it differs the print was resized

TS 11/14/18 — OPTION A — LEVEL — SCHEDULE

lots 56.4% · road 32.0% · wastage 11.6%

perpendicular down from D → P · perpendicular up from B → Q · three identical rectangles · road 20.00 ft ⊥ at its narrowest, opening to 42.49 ft

sides 72.60 / 25.00 / 63.20 / 30.00 m · closure = 1766.9 m² = 19,019 ft² = 43.66 cents · W1 = west corner triangle A-P-D, W2 = east corner triangle B-Q-C

closure: Heron(63.20, 25.00, 79.00) + Heron(79.00, 72.60, 30.00) = 678.0 + 1088.9 = 1766.9 m²

corner angles A 82.52° B 120.88° C 65.68° D 90.92° · A+C = 148.20° so the quadrilateral is NOT cyclic — Brahmagupta's 1838.5 m² is out of reach for this shape · diagonals AC 79.00 m, BD 66.34 m

SCHEDULE — every side length in feet, every area from its own arithmetic

piece	sides (ft)	area = base × height	area ft²	cents	m²
Lot 1	**64.85 × 55.10 × 64.85 × 55.10**	**64.85 × 55.10**	**3,573**	**8.20**	**331.9**
Lot 2	**64.85 × 55.10 × 64.85 × 55.10**	**64.85 × 55.10**	**3,573**	**8.20**	**331.9**
Lot 3	**64.85 × 55.10 × 64.85 × 55.10**	**64.85 × 55.10**	**3,573**	**8.20**	**331.9**
road bay 1 (over Lot 1)	64.85 × 35.04 × 65.27 × 42.49	(42.49 + 35.04)/2 × 64.85	2,513	5.77	233.5
road bay 2 (over Lot 2)	64.85 × 27.58 × 65.27 × 35.04	(35.04 + 27.58)/2 × 64.85	2,030	4.66	188.6
road bay 3 (over Lot 3)	64.85 × 20.13 × 65.27 × 27.58	(27.58 + 20.13)/2 × 64.85	1,547	3.55	143.7
LOTS + ROAD			**16,810**	**38.59**	**1561.7**
W1 (A-P-D)	12.81 × 97.59 × 98.43	½ × 98.43 × 12.70	625	1.44	58.1
W2 (B-Q-C)	75.23 × 42.37 × 82.02	½ × 82.02 × 38.61	1,583	3.64	147.1
WASTAGE			**2,209**	**5.07**	**205.2**
TOTAL			**19,019**	**43.66**	**1766.9**

TALLY 16,810 (lots + road) + 2,209 (wastage) = 19,019 ft² = 43.66 cents = 1766.9 m² = the whole plot

REGISTERED EXTENT is 1875 m² (20,182 ft² / 46.33 cents) — 108.1 m² more than these four sides can close (absolute maximum 1838.5 m²).

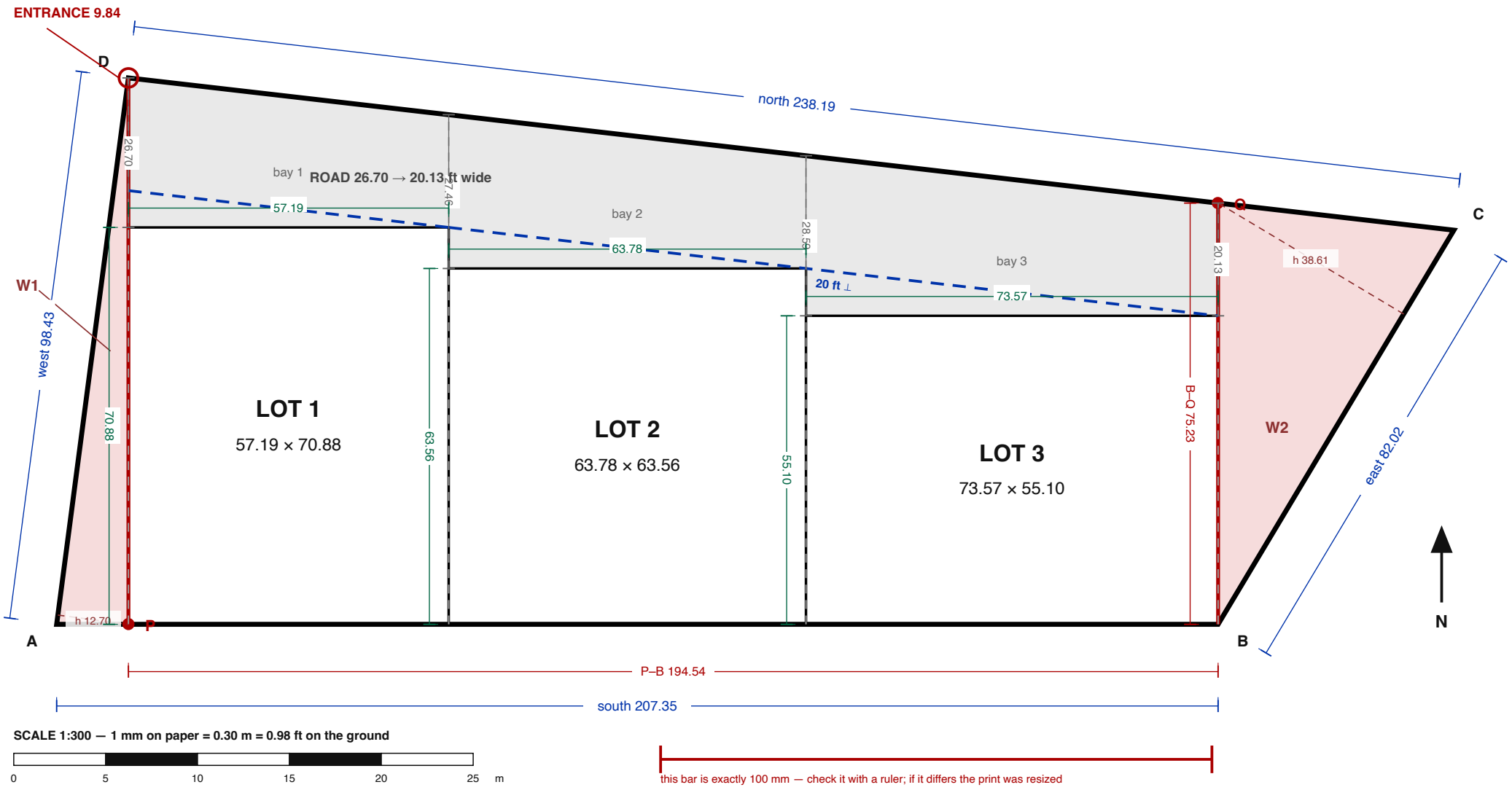
To price on the register multiply every area above by 1.0612: lot = 3,792 ft² = 8.70 cents. The fitted-sides option does this by stretching the boundary instead.

TS 11/14/18 · Ward-16 Thallakulam · Madurai North

OPTION B — STEPPED

scale 1:300 — print at 100% on A4

north up · all figures in feet



SCALE 1:300 — 1 mm on paper = 0.30 m = 0.98 ft on the ground

0 5 10 15 20 25 m

this bar is exactly 100 mm — check it with a ruler; if it differs the print was resized

TS 11/14/18 — OPTION B — STEPPED — SCHEDULE

lots 63.9% · road 24.4% · wastage 11.6%

perpendicular down from D → P · perpendicular up from B → Q · rear line steps down west→east · road 20.00 ft ⊥ at its narrowest, opening to 26.70 ft

sides 72.60 / 25.00 / 63.20 / 30.00 m · closure = 1766.9 m² = 19,019 ft² = 43.66 cents · W1 = west corner triangle A-P-D, W2 = east corner triangle B-Q-C

closure: Heron(63.20, 25.00, 79.00) + Heron(79.00, 72.60, 30.00) = 678.0 + 1088.9 = 1766.9 m²

corner angles A 82.52° B 120.88° C 65.68° D 90.92° · A+C = 148.20° so the quadrilateral is NOT cyclic — Brahmagupta's 1838.5 m² is out of reach for this shape · diagonals AC 79.00 m, BD 66.34 m

SCHEDULE — every side length in feet, every area from its own arithmetic

piece	sides (ft)	area = base × height	area ft²	cents	m²
Lot 1	**57.19 × 70.88 × 57.19 × 70.88**	**57.19 × 70.88**	**4,054**	**9.31**	**376.6**
Lot 2	**63.78 × 63.56 × 63.78 × 63.56**	**63.78 × 63.56**	**4,054**	**9.31**	**376.6**
Lot 3	**73.57 × 55.10 × 73.57 × 55.10**	**73.57 × 55.10**	**4,054**	**9.31**	**376.6**
road bay 1 (over Lot 1)	57.19 × 20.13 × 57.56 × 26.70	(26.70 + 20.13)/2 × 57.19	1,339	3.07	124.4
road bay 2 (over Lot 2)	63.78 × 20.13 × 64.20 × 27.46	(27.46 + 20.13)/2 × 63.78	1,518	3.48	141.0
road bay 3 (over Lot 3)	73.57 × 20.13 × 74.05 × 28.59	(28.59 + 20.13)/2 × 73.57	1,792	4.11	166.5
LOTS + ROAD			**16,810**	**38.59**	**1561.7**
W1 (A-P-D)	12.81 × 97.59 × 98.43	½ × 98.43 × 12.70	625	1.44	58.1
W2 (B-Q-C)	75.23 × 42.37 × 82.02	½ × 82.02 × 38.61	1,583	3.64	147.1
WASTAGE			**2,209**	**5.07**	**205.2**
TOTAL			**19,019**	**43.66**	**1766.9**

TALLY 16,810 (lots + road) + 2,209 (wastage) = 19,019 ft² = 43.66 cents = 1766.9 m² = the whole plot

REGISTERED EXTENT is 1875 m² (20,182 ft² / 46.33 cents) — 108.1 m² more than these four sides can close (absolute maximum 1838.5 m²).

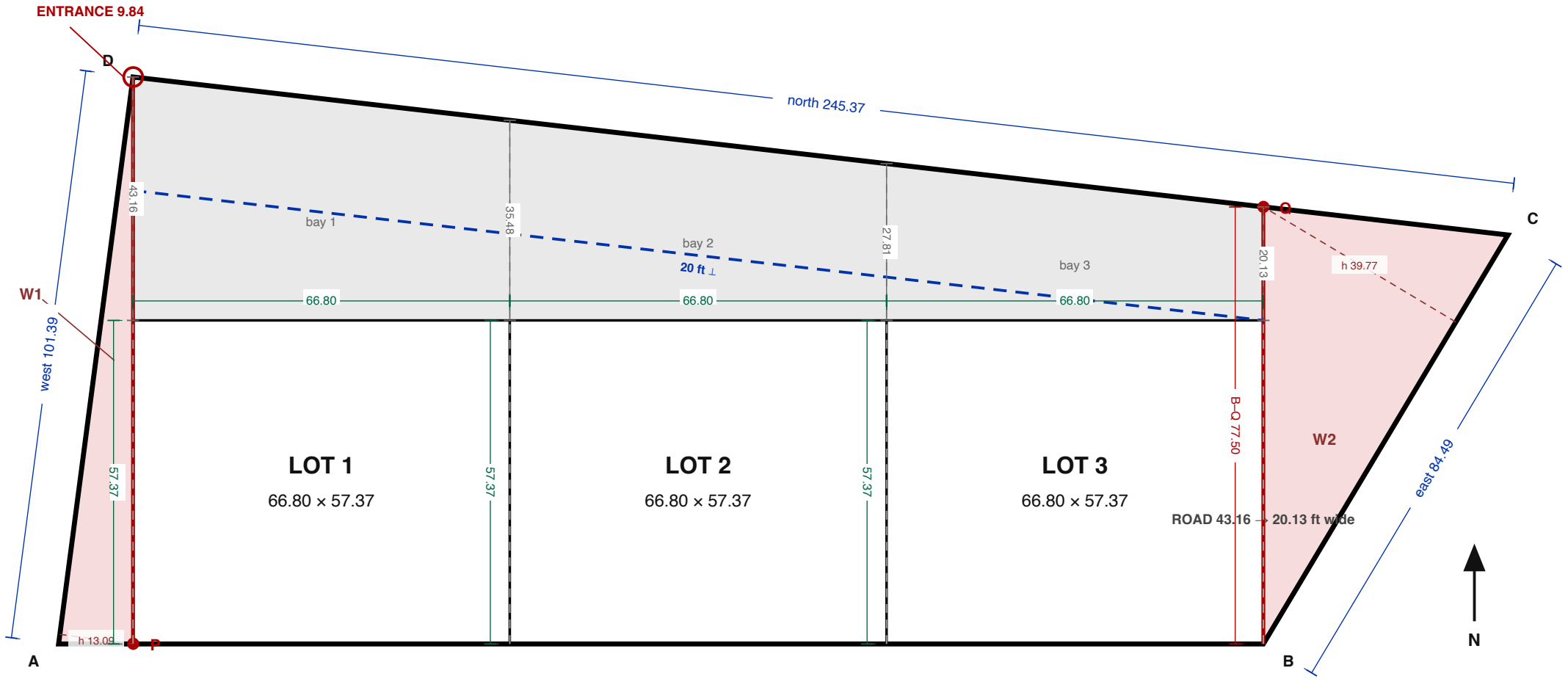
To price on the register multiply every area above by 1.0612: lot = 4,302 ft² = 9.88 cents. The fitted-sides option does this by stretching the boundary instead.

TS 11/14/18 · Ward-16 Thallakulam · Madurai North

OPTION C — LEVEL, SIDES FITTED TO 1875 m²

scale 1:300 — print at 100% on A4

north up · all figures in feet



SCALE 1:300 — 1 mm on paper = 0.30 m = 0.98 ft on the ground

0 5 10 15 20 25 m

this bar is exactly 100 mm — check it with a ruler; if it differs the print was resized

TS 11/14/18 — OPTION C — LEVEL, SIDES FITTED TO 1875 m² — SCHEDULE

lots 57.0% · road 31.4% · wastage 11.6%

perpendicular down from D → P · perpendicular up from B → Q · three identical rectangles · road 20.00 ft ⊥ at its narrowest, opening to 43.16 ft

sides 74.79 / 25.75 / 65.10 / 30.90 m · closure = 1875.0 m² = 20,182 ft² = 46.33 cents · W1 = west corner triangle A-P-D, W2 = east corner triangle B-Q-C

closure: Heron(65.10, 25.75, 81.38) + Heron(81.38, 74.79, 30.90) = 719.5 + 1155.5 = 1875.0 m²

corner angles A 82.52° B 120.88° C 65.68° D 90.92° · A+C = 148.20° so the quadrilateral is NOT cyclic — Brahmagupta's 1951.0 m² is out of reach for this shape · diagonals AC 81.38 m, BD 68.34 m

SCHEDULE — every side length in feet, every area from its own arithmetic

piece	sides (ft)	area = base × height	area ft²	cents	m²
Lot 1	**66.80 × 57.37 × 66.80 × 57.37**	**66.80 × 57.37**	**3,832**	**8.80**	**356.0**
Lot 2	**66.80 × 57.37 × 66.80 × 57.37**	**66.80 × 57.37**	**3,832**	**8.80**	**356.0**
Lot 3	**66.80 × 57.37 × 66.80 × 57.37**	**66.80 × 57.37**	**3,832**	**8.80**	**356.0**
road bay 1 (over Lot 1)	66.80 × 35.48 × 67.24 × 43.16	(43.16 + 35.48)/2 × 66.80	2,627	6.03	244.0
road bay 2 (over Lot 2)	66.80 × 27.81 × 67.24 × 35.48	(35.48 + 27.81)/2 × 66.80	2,114	4.85	196.4
road bay 3 (over Lot 3)	66.80 × 20.13 × 67.24 × 27.81	(27.81 + 20.13)/2 × 66.80	1,601	3.68	148.8
LOTS + ROAD			**17,839**	**40.95**	**1657.3**
W1 (A-P-D)	13.20 × 100.53 × 101.39	½ × 101.39 × 13.09	663	1.52	61.6
W2 (B-Q-C)	77.50 × 43.65 × 84.49	½ × 84.49 × 39.77	1,680	3.86	156.1
WASTAGE			**2,344**	**5.38**	**217.7**
TOTAL			**20,182**	**46.33**	**1875.0**

TALLY 17,839 (lots + road) + 2,344 (wastage) = 20,182 ft² = 46.33 cents = 1875.0 m² = the whole plot

SIDES STRETCHED 3.01% from the FMB's 72.60 / 25.00 / 63.20 / 30.00 m so the closure equals the REGISTERED 1875 m² exactly.

Shape and every angle unchanged: each length ×1.03014, each area ×1.0612. Verify on the ground before relying on it.